PENPOT_INTEGRATION

HelixCode PenPot Integration

This document describes the integration between HelixCode and PenPot for design system management and collaboration.

Overview

The PenPot integration provides: - **Design System Management**: Centralized design tokens and components - **Team Collaboration**: Multi-user design review and feedback - **Asset Export**: Automated export of design assets for development - **Version Control**: Design versioning and change tracking - **API Integration**: Programmatic access to design resources

Prerequisites

Required Files

- `penpot.txt` - PenPot API authentication token
- `Design/exports/` - Directory containing design assets

Required Tools

- `curl` - HTTP client for API calls
- `jq` - JSON processor for API responses

Setup

1. Obtain PenPot Token

1. Log into your PenPot account
2. Navigate to Settings → API Tokens
3. Generate a new token with appropriate permissions
4. Save the token to `penpot.txt` in the project root

2. Run Integration Setup

```bash
# Make script executable
chmod +x penpot-integration.sh
```

```
# Run complete setup
./penpot-integration.sh setup
```

3. Manual Design Import

After setup, import designs manually:

1. Access your PenPot project at the provided URL
2. Create design files and components
3. Import assets from `Design/exports/` directory
4. Set up design system with colors, typography, and components

Integration Features

Design System Components

Color Palette

- Primary, secondary, and accent colors
- Semantic color tokens
- Dark/light theme variants

Typography Scale

- Heading hierarchy (H1-H6)
- Body text variants
- Code and monospace fonts
- Responsive type scales

Component Library

- Buttons (primary, secondary, tertiary)
- Form controls (inputs, selects, checkboxes)
- Navigation elements
- Cards and containers
- Modal dialogs

Asset Management

Export Structure

```
Design/exports/
├── png/
```

```
│   ├── screens/
│   │   ├── desktop-main-workspace.png
│   │   ├── mobile-ai-chat.png
│   │   └── terminal-dashboard.png
│   └── components/
│       ├── buttons-primary.png
│       └── cards-elevated.png
├── pdf/
│   └── documentation/
│       ├── design-system.pdf
│       └── component-library.pdf
└── export-summary.json
```

Automated Exports

- Component screenshots for documentation
- Design system specifications
- Color palette exports
- Typography scales

API Integration

Available Endpoints

Project Management

- `GET /api/rpc/command/get-profile` - User profile
- `POST /api/rpc/command/create-project` - Create project
- `GET /api/rpc/command/get-projects` - List projects

File Management

- `POST /api/rpc/command/create-file` - Create design file
- `GET /api/rpc/command/get-file` - Get file data
- `POST /api/rpc/command/update-file` - Update file

Asset Management

- `POST /api/rpc/command/upload-file-media-object` - Upload assets
- `GET /api/rpc/command/get-file-media-objects` - List assets

Example API Usage

```bash
# Test connection
./penpot-integration.sh test

# Create project
curl -X POST \
  -H "Authorization: Token $(cat penpot.txt)" \
  -H "Content-Type: application/json" \
  -d '{"type": "create-project", "name": "HelixCode", "team-id":
        "default"}' \
  https://design.penpot.app/api/rpc/command/create-project
```

Workflow Integration

Development Workflow

1. **Design Creation**: Designers create components in PenPot
2. **Review & Approval**: Team reviews and approves designs
3. **Asset Export**: Automated export of approved designs
4. **Implementation**: Developers implement using exported assets
5. **Validation**: Design-system compliance validation

Collaboration Features

Team Management

- Role-based access control
- Design review workflows
- Comment and feedback system
- Version history and rollback

Design Tokens

- Centralized token management
- Automated code generation
- Cross-platform consistency
- Theme variant support

Configuration

Environment Variables

```bash
# PenPot instance URL (optional)
PENPOT_BASE_URL=https://design.penpot.app

# Project settings (set via API)
PENPOT_PROJECT_NAME="HelixCode Design System"
PENPOT_TEAM_ID="default"
```

Integration Settings

Design System Configuration

```json
{
  "colors": {
    "primary": "#0066FF",
    "secondary": "#666666",
    "accent": "#FF3366"
  },
  "typography": {
    "fontFamily": "Inter, system-ui, sans-serif",
    "scale": [12, 14, 16, 18, 20, 24, 30, 36, 48]
  },
  "spacing": {
    "unit": 8,
    "scale": [0, 4, 8, 16, 24, 32, 48, 64, 96]
  }
}
```

Usage Examples

Command Line Interface

```bash
# Check integration status
./penpot-integration.sh status

# Test API connection
./penpot-integration.sh test

# Setup complete integration
./penpot-integration.sh setup
```

Manual Operations

Import Design Assets

1. Open PenPot web interface
2. Navigate to your project
3. Use "Import" feature to add design files
4. Organize assets in libraries

Export Development Assets

1. Select components for export
2. Choose export format (SVG, PNG, PDF)
3. Download to `Design/exports/` directory
4. Update export manifest

Troubleshooting

Common Issues

Authentication Errors

- Verify token in `penpot.txt` is valid
- Check token permissions in PenPot settings
- Ensure token hasn't expired

API Connection Issues

- Verify network connectivity
- Check PenPot service status
- Validate API endpoint URLs

Asset Import Issues

- Check file formats (SVG, PNG, PDF supported)
- Verify file permissions
- Ensure adequate storage space

Debug Mode

Enable debug output for troubleshooting:

```
# Add debug flag to see detailed output
PENPOT_DEBUG=1 ./penpot-integration.sh setup
```

Security Considerations

Token Security

- Store `penpot.txt` securely
- Never commit tokens to version control
- Use environment variables in production
- Rotate tokens regularly

Access Control

- Limit token permissions to minimum required
- Use team-based access controls
- Audit access regularly
- Monitor for unauthorized usage

Best Practices

Design System Management

1. **Centralize Tokens**: Use design tokens for consistency
2. **Version Control**: Track design system changes
3. **Documentation**: Maintain design system documentation
4. **Testing**: Validate design implementation

Collaboration Workflow

1. **Clear Roles**: Define designer and developer responsibilities
2. **Review Process**: Establish design review workflows
3. **Feedback Loops**: Implement efficient feedback mechanisms
4. **Quality Gates**: Set quality standards for design assets

Integration Maintenance

1. **Regular Updates**: Keep integration scripts current
2. **API Monitoring**: Monitor for API changes
3. **Error Handling**: Implement robust error handling
4. **Backup Strategy**: Backup design assets and configurations

Support

Resources

- PenPot Documentation
- PenPot API Reference
- Design System Best Practices

Getting Help

1. Check integration status: `./penpot-integration.sh status`
2. Test connection: `./penpot-integration.sh test`
3. Review logs for error messages
4. Consult PenPot community forums

Future Enhancements

Planned Features

- Automated design token generation
- Real-time design-dev synchronization
- Component code generation
- Design system analytics
- Multi-platform asset optimization

Integration Roadmap

1. **Phase 1**: Basic API integration and asset management
2. **Phase 2**: Automated design token synchronization
3. **Phase 3**: Real-time collaboration features
4. **Phase 4**: Advanced analytics and optimization

This integration establishes a robust foundation for design-development collaboration, ensuring consistency and efficiency across the HelixCode project.